

C Programming — Arrays

Theory Questions & Programming Problems | Intern Training Session

Section A: Theory Questions

- 1 What is an array? Why do we need arrays?
- 2 How is an array different from a normal variable?
- 3 What is the difference between an array and a linked list (briefly)?
- 4 How are array elements stored in memory?
- 5 What is the index of an array? Why does indexing start from 0 in C?
- 6 What is the difference between declaration and initialization of an array?
- 7 What happens if you access an array index out of bounds in C?
- 8 What is the size of an array, and how can you calculate it using sizeof?
- 9 What is a multidimensional array? Give an example.
- 10 How are 2D arrays stored in memory (row-major order)?
- 11 What is the relationship between arrays and pointers in C?
- 12 How do you pass an array to a function? Why can't you pass it "by value" like a normal variable?
- 13 What is the difference between a static array and a dynamic array?
- 14 Can the size of an array be changed after declaration? What's the alternative?
- 15 What is an array of strings (array of character arrays)?
- 16 What is the difference between `char arr[10]` and `char *arr`?
- 17 What is a jagged array, and does C support it directly?
- 18 What is the difference between linear search and binary search on arrays?
- 19 Why must an array be sorted before applying binary search?
- 20 What is array decay, and what does it mean when an array is passed to a function?

Section B: Programming Problems

Basic

- 1 Write a program to input and print elements of an array.
- 2 Write a program to find the sum and average of array elements.
- 3 Write a program to find the largest and smallest element in an array.
- 4 Write a program to count even and odd numbers in an array.
- 5 Write a program to reverse an array.

6 Write a program to copy elements of one array into another.

Intermediate

7 Write a program to search an element using linear search.

8 Write a program to search an element using binary search.

9 Write a program to sort an array using bubble sort.

10 Write a program to sort an array using selection sort.

11 Write a program to find the second largest element in an array.

12 Write a program to remove duplicate elements from an array.

13 Write a program to merge two arrays into a third array.

14 Write a program to find the frequency of each element in an array.

15 Write a program to left-rotate / right-rotate an array by one position.

16 Write a program to check if an array is a palindrome.

2D Arrays / Matrices

17 Write a program to input and print a 2D array (matrix).

18 Write a program to add two matrices.

19 Write a program to multiply two matrices.

20 Write a program to find the transpose of a matrix.

21 Write a program to check if a matrix is symmetric.

22 Write a program to find the sum of diagonal elements of a square matrix.

Advanced

23 Write a program to find the union and intersection of two arrays.

24 Write a program to find the missing number in an array of 1 to N.

25 Write a program to find the maximum subarray sum (basic approach).

26 Write a program using functions and pointers to pass an array and modify it inside the function.